

ASSIGNMENT 17

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ROLL NO : CSB24021

```
package com.example.registration;

public class InvalidEmailException
    extends Exception {

    public InvalidEmailException(
        String message
    ) {

        super(message);
    }
}
```

UnderageException.java

```
package com.example.registration;

public class UnderageException
    extends RuntimeException {

    public UnderageException(
        String message
    ) {

        super(message);
    }
}
```

RegistrationService.java

```
package com.example.registration;

import java.util.regex.Pattern;

public class RegistrationService {

    private static final String EMAIL_REGEX =
```

```

        "^[A-Za-z0-9+_.-]+@[A-Za-z0-9.-]+$";

private static final Pattern pattern =
    Pattern.compile(EMAIL_REGEX);

public boolean registerUser(
    String email,
    int age
) throws InvalidEmailException {

    assert age >= 0 :
        "Age cannot be negative";

    if (email == null ||
        email.trim().isEmpty()) {

        throw new InvalidEmailException(
            "Email cannot be null or empty"
        );
    }

    if (!pattern.matcher(email)
        .matches()) {

        throw new InvalidEmailException(
            "Invalid email format"
        );
    }

    if (age < 18) {

        throw new UnderageException(
            "User must be at least 18 years old"
        );
    }

    return true;
}
}

```

RegistrationServiceTest.java

```
package com.example.registration;

import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;

import static org.junit.jupiter.api.Assertions.*;

public class RegistrationServiceTest {

    private RegistrationService service;

    @BeforeEach
    void setup() {

        service =
            new RegistrationService();
    }

    @Test
    void testSuccessfulRegistration()
        throws InvalidEmailException {

        boolean result =
            service.registerUser(
                "user@example.com",
                21
            );

        assertTrue(result);
    }

    @Test
    void testInvalidEmail() {

        assertThrows(
            InvalidEmailException.class,
            () -> {
```

```

        service.registerUser(
            "invalid-email",
            22
        );
    }

};

@Test
void testUnderageUser() {

    assertThrows(
        UnderageException.class,
        () -> {

            service.registerUser(
                "user@example.com",
                16
            );
        }
    );
}
}

```

OUTPUT:

T E S T S

Running com.example.registration.RegistrationServiceTest

Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.087 s --
ationServiceTest

Results:

Tests run: 1, Failures: 0, Errors: 0, Skipped: 0

BUILD SUCCESS

Total time: 2.894 s

Finished at: 2026-05-21T21:09:01+05:30
